

PREDIX HER2 — patients with complete data across all five modalities

n = 110 (46 pCR)

DHP n = 59 (24 pCR)

T-DM1 n = 51 (22 pCR)

every modality-specific model is trained and evaluated on this cohort alone: within each repeat it is fitted on the outer training folds and applied to the patients held out from them

Candidate features — a-priori biological curation, no outcome used

Clinical
5 metrics

Transcriptomic
42 metrics

Genomic
41 metrics

Proteomic
19 metrics

WSI
3 metrics

110 pre-defined metrics → fixed outcome-blind deduplication of co-amplified and near-identical features (~18) → 92 candidates

→ per-scenario, within-modality deduplication before any split: $|r| \geq 0.80$, exact agreement ≥ 0.80 (segment-called CNA), $\kappa \geq 0.80$ (binary) → ~11 pooled / ~12 DHP / ~6 T-DM1

OUTER LOOP — stratified 5-fold cross-validation, RepeatedStratifiedKfold, seed 42

training

training

training

held-out
test fold

training

each fold is held out in turn (~80 % train / ~20 % test), stratified on pCR

the whole partition is redrawn × 200 repeats = 1,000 outer evaluations (pooled cohort)

arm-specific models: 5-fold × 100 repeats = 500 evaluations per arm,
with 3-fold inner cross-validation. No shuffle-split resampling at any stage.

WITHIN ONE OUTER TRAINING FOLD — fitted on training patients only

Preprocessing, in this order

near-zero-variance filter
median imputation, then standardisation
univariate screen: Mann–Whitney AUROC,
BH $q \leq 0.25$, keep 5–40 candidates
(clinical, WSI: ≤ 6 candidates, unscreened)
no in-fold correlation filter: redundancy is removed once, above, before any fold is drawn

Inner 5-fold cross-validation

Stage A the five classifier families below are compared under fixed grids; importances become percentile ranks and define the feature signature, capped at ≥ 5 pCR events per selected variable
Stage B the winning family is tuned by grid search, restricted to the discovered signature

ElasticNet logistic

RandomForest

ExtraTrees

HistGradBoost

SVM RBF

SVM Linear

probabilities recalibrated by Platt scaling fitted inside the cross-validation

→ a leakage-safe out-of-fold probability for each of the five modalities

Late-fusion stacking

elastic-net logistic regression (L1 + L2, $l1_ratio = 0.5$) over the five calibrated probability streams;
the L1 term sets non-contributing modalities to exactly zero, giving interpretable modality weights

Applied unchanged to the held-out outer test fold

in every repeat each patient receives exactly one out-of-fold predicted probability;
no patient is ever scored by a model that saw them in training

Locked consensus model — aggregated over the 1,000 outer folds

for each modality the locked classifier is the family that won the most outer folds;

the signature is aggregated only over the folds that family won — one classifier, one matching signature

K = the median per-fold signature size; correlated DNA and RNA candidates are pooled to one representative

Reported performance

the metric (AUROC, AUPRC, Brier) is computed on each repeat's complete out-of-fold vector and averaged over the 200 repeats (100 per arm); 95 % confidence intervals are patient-level cluster bootstraps — 2,000 stratified resamples of patients, a resampled patient carrying all of its repeat predictions. Predictions are never averaged across repeats or across models.